

AR0234CS Register Reference

AND9812/D

Introduction

This reference document describes the AR0234CSSR registers. Summary and detailed information are presented in separate sections:

- “Register Lists and Default Values” on page 4
- “Register Descriptions” on page 15

Note: Throughout this document, green1 corresponds to greenR; green2 corresponds to greenB.

Conventions and Notations

This document follows the conventions and notations described below.

- Hexadecimal numbers have a 0x prefix
 - Binary numbers have 0b prefix
- Example: 0b1010 = 0xA

Register Address Space

The AR0234CS provides a 16-bit register address space accessed through a serial interface. Each register location is 8 or 16 bits in size.

The address space is divided into the five major regions shown in Table 1.

Table 1. ADDRESS SPACE REGIONS

Address Range	Description
0x0000–0x0FFF	Reserved
0x1000–0x1FFF	Reserved
0x2000–0x2FFF	Manufacturer-specific registers (read-only and read-write dynamic registers)
0x3000–0x3FFF	Manufacturer-specific registers (read-only and read-write dynamic registers)
0x4000–0xFFFF	Reserved (undefined)

Register Notation

The underlying mechanism for reading and writing registers provides byte write capability. However, it is convenient to consider some registers as multiple adjacent bytes. The AR0234CS uses 8-bit, 16-bit, and 32-bit registers, all implemented as 1 or more bytes at naturally aligned, contiguous locations in the address space.

In this document, registers are described either by address or by name. When registers are described by address, the size of the registers is explicit. For example, R0x301C is an 8-bit register at address 0x301C, and R0x3000–1 is a 16-bit register at address 0x3000–0x3001. When registers are described by name, the size of the register is implicit. It is necessary to refer to the register table to determine that model_id is a 16-bit register.

Register Aliases

A consequence of the internal architecture of the AR0234CS is that some registers are decoded at multiple addresses. Some registers in “configuration space” are also decoded in “manufacturer-specific space.” To provide unique names for all registers, the name of the register within manufacturer-specific register space has a trailing underscore. For example, R0x0000–1 is model_id, and R0x3000–1 is model_id_. The effect of reading or writing a register through any of its aliases is identical.

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Bit Fields

Some registers provide control of several different pieces of related functionality, and this makes it necessary to refer to bit fields within registers. As an example of the notation used for this, the least significant 4 bits of the model_id register are referred to as model_id[3:0] or R0x0000-1[3:0].

Bit Field Aliases

In addition to the register aliases described above, some register fields are aliased in multiple places. For example, R0x301C (mode_select) only has one operational bit, R0x301C[0]. This bit is aliased to R0x301A-B[2]. The effect of reading or writing a bit field through any of its aliases is identical.

Byte Ordering

Registers that occupy more than 1 byte of address space are shown with the lowest address in the highest-order byte lane to match the byte-ordering on the bus. For example, the model_id register is R0x3000-1. In the register table the default value is shown as 0x0856. This means that a READ from address 0x3000 would return 0x08, and a READ from

address 0x3001 would return 0x56. When reading this register as two 8-bit transfers on the serial interface, the 0x08 will appear on the serial interface first, followed by the 0x56.

Address Alignment

All register addresses are aligned naturally. Registers that occupy two bytes of address space are aligned to even 16-bit addresses, and registers that occupy four bytes of address space are aligned to 16-bit addresses that are an integer multiple of 4.

Bit Representation

For clarity, 32-bit hex numbers are shown with an underscore between the upper and lower 16 bits. For example: 0x3000_01AB.

Data Format

Most registers represent an unsigned binary value or set of bit fields. For all other register formats, the format is stated explicitly at the start of the register description. The notation for these formats is shown in Table 2.

Table 2. DATA FORMATS

Name	Description
FIX16	Signed fixed-point, 16-bit number: two's complement number, 8 fractional bits. Examples: 0x0100 = 1.0, 0x8000 = -128, 0xFFFF = -0.0039065
UFIX16	Unsigned fixed-point, 16-bit number: 8.8 format. Examples: 0x0100 = 1.0, 0x280 = 2.5
FLP32	Signed floating-point, 32-bit number: IEEE 754 format. Example: 0x4280_0000 = 64.0

Register Behavior

Registers vary from "read-only", "write-only", "read/write", and "read, write-1-to-clear".

DOUBLE-BUFFERED REGISTERS

Some sensor settings cannot be changed during frame readout. For example, changing x_addr_start partway through frame readout would result in inconsistent row lengths within a frame. To avoid this, the AR0234CS double-buffers many registers by implementing a "pending" and a "live" version. READs and WRITEs access the pending register; the live register controls the sensor operation.

The value in the pending register is transferred to a live register at a fixed point in the frame timing, called frame start. Frame start is defined as the point at which the first dark row is read out internally to the sensor. In the register tables the "Buffering" column shows which registers or register fields are single- or double-buffered

Buffering

In register tables, buffering shows the timing with which a newly-written register value takes effect. The notation used is:

Blank – Unbuffered. By default register update takes effect immediately.

S – Single frame sync'd. Register update in frame N takes effect in frame N+1.

D – Double frame sync'd. Register update in frame N takes effect in frame N+2.

Bad Frames

A bad frame is a frame where all rows do not have the same integration time or where offsets to the pixel values have changed during the frame.

Many changes to the sensor register settings can cause a bad frame. For example, when line_length_pck is changed, the new register value does not affect sensor behavior until the next frame start. However, the frame that would be read out at that frame start will have been integrated using the old row width, so reading it out using the new row width would result in a frame with an incorrect integration time.

By default, bad frames are not masked. If the masked bad frame option is enabled, both LV and FV are inhibited for these frames so that the vertical blanking time between frames is extended by the frame time.

In the register tables, the “Bad Frame” column shows where changing a register or register field will cause a bad frame. This notation is used:

Blank – No. Changing the register value will not produce a bad frame.

Y – Yes. Changing the register value might produce a bad frame.

YM – Yes. But the bad frame will be masked out when mask_corrupted_frames (R0x301A[9]) is set to “1”.

Embedded

In register tables, the embedded column notes whether or not the register is present in the per – frame embedded data.

The notation used is:

Blank – By default, a register is not present in the embedded data

E – The register is present in the embedded data.

Locked

In register tables, locked notes whether writes to the register are protected by R0x3010. The notation used is

Blank – By default, writes to a register are not protected by R0x3010

L – Writes to the register are protected by R0x3010.

Table 3. MANUFACTURER SPECIFIC

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x3000	CHIP_VERSION_REG	dddd dddd dddd dddd	2646 (0x0A56)
R0x3002	Y_ADDR_START	0000 0ddd dddd dddd	8 (0x0008)
R0x3004	X_ADDR_START	0000 dddd dddd dddd	8 (0x0008)
R0x3006	Y_ADDR_END	0000 0ddd dddd dddd	1207 (0x04B7)
R0x3008	X_ADDR_END	0000 dddd dddd dddd	1927 (0x0787)
R0x300A	FRAME_LENGTH_LINES	dddd dddd dddd dddd	1216 (0x04C0)
R0x300C	LINE_LENGTH_PCK	dddd dddd dddd dddd	612 (0x0264)
R0x300E	REVISION_NUMBER	dddd dddd dddd dddd	0 (0x0000)
R0x3010	LOCK_CONTROL	dddd dddd dddd dddd	48879 (0xBEEF)
R0x3012	COARSE_INTEGRATION_TIME	dddd dddd dddd dddd	16 (0x0010)
R0x3014	FINE_INTEGRATION_TIME	dddd dddd dddd dddd	10 (0x000A)
R0x3016	COARSE_INTEGRATION_TIME_CB	dddd dddd dddd dddd	16 (0x0010)
R0x3018	FINE_INTEGRATION_TIME_CB	dddd dddd dddd dddd	0 (0x0000)
R0x301A	RESET_REGISTER	d0dd dddd dddd dddd	8280 (0x2058)
R0x301C	MODE_SELECT_	0000 000d	0 (0x00)
R0x301D	IMAGE_ORIENTATION_	0000 00dd	0 (0x00)
R0x301E	DATA_PEDESTAL	0000 00dd dddd dddd	42 (0x002A)
R0x3021	SOFTWARE_RESET_	0000 000d	0 (0x00)
R0x3022	GROUPED_PARAMETER_HOLD_	0000 000d	0 (0x00)
R0x3023	MASK_CORRUPTED_FRAMES_	0000 000d	0 (0x00)
R0x3024	PIXEL_ORDER_	0000 00??	0 (0x00)
R0x3026	GPI_STATUS	???? ???? ???? ???? ?	25856 (0x6500)
R0x3028	ROW_SPEED	0000 0000 0ddd 0000	16 (0x0010)
R0x302A	VT_PIX_CLK_DIV	0000 0000 000d dddd	5 (0x0005)
R0x302C	VT_SYS_CLK_DIV	0000 0000 000d dddd	1 (0x0001)

Table 3. MANUFACTURER SPECIFIC (continued)

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x302E	PRE_PLL_CLK_DIV	0000 0000 00dd dddd	3 (0x0003)
R0x3030	PLL_MULTIPLIER	0000 0000 dddd dddd	50 (0x0032)
R0x3034	CTX_CONTROL_REG	dd00 0ddd dddd dddd	0 (0x0000)
R0x3036	OP_PIX_CLK_DIV	0000 0000 000d dddd	10 (0x000A)
R0x3038	OP_SYS_CLK_DIV	0000 0000 000d dddd	1 (0x0001)
R0x303A	FRAME_COUNT	???? ???? ???? ????	65535 (0xFFFF)
R0x303C	FRAME_STATUS	0000 0000 0000 ????	0 (0x0000)
R0x303E	LINE_LENGTH_PCK_CB	dddd dddd dddd dddd	612 (0x0264)
R0x3040	READ_MODE	dddd dddd ddd0 0000	0 (0x0000)
R0x3042	EXTRA_DELAY	dddd dddd dddd dddd	0 (0x0000)
R0x3056	GREEN1_GAIN	0000 0ddd dddd dddd	128 (0x0080)
R0x3058	BLUE_GAIN	0000 0ddd dddd dddd	128 (0x0080)
R0x305A	RED_GAIN	0000 0ddd dddd dddd	128 (0x0080)
R0x305C	GREEN2_GAIN	0000 0ddd dddd dddd	128 (0x0080)
R0x305E	GLOBAL_GAIN	0000 0ddd dddd dddd	128 (0x0080)
R0x3060	ANALOG_GAIN	0ddd dddd 0ddd dddd	14 (0x000E)
R0x3064	SMIA_TEST	000d dddd d0d0 dddd	6146 (0x1802)
R0x3066	CTX_WR_DATA_REG	dddd dddd dddd dddd	0 (0x0000)
R0x3068	CTX_RD_DATA_REG	dddd dddd dddd dddd	0 (0x0000)
R0x306E	DATAPATH_SELECT	dddd dddd 000d dddd	23568 (0x5C10)
R0x3070	TEST_PATTERN_MODE	0000 000d 0000 0ddd	0 (0x0000)
R0x3072	TEST_DATA_RED	0000 00dd dddd dddd	0 (0x0000)
R0x3074	TEST_DATA_GREENR	0000 00dd dddd dddd	0 (0x0000)
R0x3076	TEST_DATA_BLUE	0000 00dd dddd dddd	0 (0x0000)
R0x3078	TEST_DATA_GREENB	0000 00dd dddd dddd	0 (0x0000)

Table 3. MANUFACTURER SPECIFIC (continued)

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x3082	OPERATION_MODE_CTRL	0000 0000 0000 dddd	3 (0x0003)
R0x3084	OPERATION_MODE_CTRL_CB	0000 0000 0000 dddd	3 (0x0003)
R0x3086	SEQ_DATA_PORT	dddd dddd dddd dddd	0 (0x0000)
R0x3088	SEQ_CTRL_PORT	?d00 00dd dddd dddd	49152 (0xC000)
R0x308A	X_ADDR_START_CB	0000 dddd dddd dddd	8 (0x0008)
R0x308C	Y_ADDR_START_CB	0000 0ddd dddd dddd	8 (0x0008)
R0x308E	X_ADDR_END_CB	0000 dddd dddd dddd	1927 (0x0787)
R0x3090	Y_ADDR_END_CB	0000 0ddd dddd dddd	1207 (0x04B7)
R0x30A0	X_EVEN_INC	0000 0000 0000 000?	1 (0x0001)
R0x30A2	X_ODD_INC	0000 0000 000d dddd	1 (0x0001)
R0x30A4	Y_EVEN_INC	0000 0000 0000 000?	1 (0x0001)
R0x30A6	Y_ODD_INC	0000 0000 0ddd dddd	1 (0x0001)
R0x30A8	Y_ODD_INC_CB	0000 0000 0ddd dddd	1 (0x0001)
R0x30AA	FRAME_LENGTH_LINES_CB	dddd dddd dddd dddd	1216 (0x04C0)
R0x30AE	X_ODD_INC_CB	0000 0000 000d dddd	3 (0x0003)
R0x30B0	DIGITAL_TEST	dddd dddd dddd dddd	40 (0x0028)
R0x30B2	TEMPSENS_DATA_REG	0000 00dd dddd dddd	0 (0x0000)
R0x30B4	TEMPSENS_CTRL_REG	dddd dddd dddd dddd	0 (0x0000)
R0x30BC	GREEN1_GAIN_CB	0000 0ddd dddd dddd	128 (0x0080)
R0x30BE	BLUE_GAIN_CB	0000 0ddd dddd dddd	128 (0x0080)
R0x30C0	RED_GAIN_CB	0000 0ddd dddd dddd	128 (0x0080)
R0x30C2	GREEN2_GAIN_CB	0000 0ddd dddd dddd	128 (0x0080)
R0x30C4	GLOBAL_GAIN_CB	0000 0ddd dddd dddd	128 (0x0080)
R0x30C6	TEMPSENS_CALIB1	dddd dddd dddd dddd	0 (0x0000)
R0x30C8	TEMPSENS_CALIB2	dddd dddd dddd dddd	0 (0x0000)

Table 3. MANUFACTURER SPECIFIC (continued)

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x30CE	GRR_CONTROL1	0000 00dd dddd 0d0d	0 (0x0000)
R0x30FE	NOISE_PEDESTAL	0000 00dd dddd dddd	42 (0x002A)
R0x3100	AECTRLREG	0000 d0dd dddd dddd	0 (0x0000)
R0x3102	AE_LUMA_TARGET_REG	dddd dddd dddd dddd	1280 (0x0500)
R0x3108	AE_MIN_EV_STEP_REG	dddd dddd dddd dddd	112 (0x0070)
R0x310A	AE_MAX_EV_STEP_REG	dddd dddd dddd dddd	8 (0x0008)
R0x310C	AE_DAMP_OFFSET_REG	dddd dddd dddd dddd	512 (0x0200)
R0x310E	AE_DAMP_GAIN_REG	dddd dddd dddd dddd	8192 (0x2000)
R0x3110	AE_DAMP_MAX_REG	dddd dddd dddd dddd	320 (0x0140)
R0x311C	AE_MAX_EXPOSURE_REG	dddd dddd dddd dddd	672 (0x02A0)
R0x311E	AE_MIN_EXPOSURE_REG	dddd dddd dddd dddd	1 (0x0001)
R0x3124	AE_DARK_CUR_THRESH_REG	dddd dddd dddd dddd	32767 (0x7FFF)
R0x312A	AE_CURRENT_GAINS	00?? ???? ???? ????	128 (0x0080)
R0x312E	ACTUAL_FRAME_LENGTH_LINES	???? ???? ???? ????	0 (0x0000)
R0x3140	AE_ROI_X_START_OFFSET	0000 dddd dddd ddd0	0 (0x0000)
R0x3142	AE_ROI_Y_START_OFFSET	0000 0ddd dddd ddd0	0 (0x0000)
R0x3144	AE_ROI_X_SIZE	0000 dddd dddd ddd0	1920 (0x0780)
R0x3146	AE_ROI_Y_SIZE	0000 0ddd dddd ddd0	1200 (0x04B0)
R0x3148	AE_HIST_BEGIN_PERC	dddd dddd dddd dddd	976 (0x03D0)
R0x314A	AE_HIST_END_PERC	dddd dddd dddd dddd	65535 (0xFFFF)
R0x314C	AE_HIST_DIV	dddd dddd dddd dddd	256 (0x0100)
R0x314E	AE_NORM_WIDTH_MIN	dddd dddd dddd dddd	32 (0x0020)
R0x3150	AE_MEAN_H	0000 0000 0000 ????	0 (0x0000)
R0x3152	AE_MEAN_L	???? ???? ???? ????	0 (0x0000)
R0x3154	AE_HIST_BEGIN_H	0000 0000 0000 ????	0 (0x0000)

Table 3. MANUFACTURER SPECIFIC (continued)

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x3156	AE_HIST_BEGIN_L	???? ???? ???? ????	0 (0x0000)
R0x3158	AE_HIST_END_H	0000 0000 0000 ????	0 (0x0000)
R0x315A	AE_HIST_END_L	???? ???? ???? ????	0 (0x0000)
R0x315C	AE_LOW_END_MEAN_H	0000 0000 0000 ????	0 (0x0000)
R0x315E	AE_LOW_END_MEAN_L	???? ???? ???? ????	0 (0x0000)
R0x3160	AE_PERC_LOW_END	???? ???? ???? ????	0 (0x0000)
R0x3162	AE_NORM_ABS_DEV	???? ???? ???? ????	0 (0x0000)
R0x3164	AE_COARSE_INTEGRATION_TIME	???? ???? ???? ????	1 (0x0001)
R0x3166	AE_AG_EXPOSURE_HI	dddd dddd dddd dddd	986 (0x03DA)
R0x3168	AE_AG_EXPOSURE_LO	dddd dddd dddd dddd	419 (0x01A3)
R0x3180	DELTA_DK_CONTROL	dddd ddd0 dddd dddd	33359 (0x824F)
R0x31AC	DATA_FORMAT_BITS	dddd dddd dddd dddd	2570 (0x0A0A)
R0x31AE	SERIAL_FORMAT	0000 00dd 0000 0ddd	516 (0x0204)
R0x31B0	FRAME_PREAMBLE	0000 0000 dddd dddd	112 (0x0070)
R0x31B2	LINE_PREAMBLE	0000 0000 dddd dddd	81 (0x0051)
R0x31B4	MIPI_TIMING_0	dddd dddd dddd dddd	21064 (0x5248)
R0x31B6	MIPI_TIMING_1	dddd dddd dddd dddd	12952 (0x3298)
R0x31B8	MIPI_TIMING_2	dddd dddd dddd dddd	36939 (0x904B)
R0x31BA	MIPI_TIMING_3	dddd dddd dddd dddd	780 (0x030C)
R0x31BC	MIPI_TIMING_4	dddd dddd dddd dddd	36490 (0x8E8A)
R0x31BE	SERIAL_CONFIG_STATUS	0000 dddd dddd dddd	35 (0x0023)
R0x31C6	SERIAL_CONTROL_STATUS	??dd dddd dddd dddd	32768 (0x8000)
R0x31C8	SERIAL_CRC_0	???? ???? ???? ????	65535 (0xFFFF)
R0x31D0	COMPANDING	0000 0000 000d d00d	0 (0x0000)
R0x31D2	STAT_FRAME_ID	dddd dddd dddd dddd	0 (0x0000)

Table 3. MANUFACTURER SPECIFIC (continued)

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x31D6	I2C_WRT_CHECKSUM	dddd dddd dddd dddd	65535 (0xFFFF)
R0x31D8	SERIAL_TEST	0000 dddd dddd dddd	0 (0x0000)
R0x31E0	PIX_DEF_ID	dd00 0000 0000 dddd	0 (0x0000)
R0x31E8	HORIZONTAL_CURSOR_POSITION	0000 0ddd dddd dddd	0 (0x0000)
R0x31EA	VERTICAL_CURSOR_POSITION	0000 dddd dddd dddd	0 (0x0000)
R0x31EC	HORIZONTAL_CURSOR_WIDTH	0000 0ddd dddd dddd	0 (0x0000)
R0x31EE	VERTICAL_CURSOR_WIDTH	0000 dddd dddd dddd	0 (0x0000)
R0x31F0	AE_STATS_CONTROL	0000 0000 0000 dddd	0 (0x0000)
R0x31FC	CCI_IDS	dddd dddd dddd dddd	12320 (0x3020)
R0x31FE	CUSTOMER_REV	???? ???? ???? ????	0000 (0x0000)
R0x3240	AE_X1_START_OFFSET	0000 dddd dddd ddd0	256 (0x0100)
R0x3242	AE_X2_START_OFFSET	0000 dddd dddd ddd0	512 (0x0200)
R0x3244	AE_X3_START_OFFSET	0000 dddd dddd ddd0	768 (0x0300)
R0x3246	AE_X4_START_OFFSET	0000 dddd dddd ddd0	1024 (0x0400)
R0x3248	AE_Y1_START_OFFSET	0000 0ddd dddd ddd0	160 (0x00A0)
R0x324A	AE_Y2_START_OFFSET	0000 0ddd dddd ddd0	320 (0x0140)
R0x324C	AE_Y3_START_OFFSET	0000 0ddd dddd ddd0	480 (0x01E0)
R0x324E	AE_Y4_START_OFFSET	0000 0ddd dddd ddd0	640 (0x0280)
R0x3250	AE_GRID_SEL_LO	dddd dddd dddd dddd	65535 (0xFFFF)
R0x3252	AE_GRID_SEL_HI	0000 000d dddd dddd	511 (0x01FF)
R0x3270	LED_FLASH_CONTROL	0000 000d dddd dddd	0 (0x0000)
R0x3338	MIPI_TEST_CNTRL	0000 0000 0000 00dd	0 (0x0000)
R0x333A	MIPI_COMPRESS_8_DATA_TYPE	00dd dddd 00dd dddd	12849 (0x3231)
R0x333C	MIPI_COMPRESS_7_DATA_TYPE	00dd dddd 00dd dddd	13620 (0x3534)
R0x333E	MIPI_COMPRESS_6_DATA_TYPE	00dd dddd 00dd dddd	14134 (0x3736)

Table 3. MANUFACTURER SPECIFIC (continued)

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x3340	MIPI_JPEG_PN9_DATA_TYPE	00dd dddd 00dd dddd	13104 (0x3330)
R0x3354	MIPI_CNTRL	0000 000d dddd dddd	43 (0x002B)
R0x3356	MIPI_TEST_PATTERN_CNTRL	0000 0000 00dd dddd	0 (0x0000)
R0x3358	MIPI_TEST_PATTERN_STATUS	0000 0000 00?? ????	63 (0x003F)
R0x3600	P_GR_P0Q0	dddd dddd dddd dddd	0 (0x0000)
R0x3602	P_GR_P0Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3604	P_GR_P0Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3606	P_GR_P0Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3608	P_GR_P0Q4	dddd dddd dddd dddd	0 (0x0000)
R0x360A	P_RD_P0Q0	dddd dddd dddd dddd	0 (0x0000)
R0x360C	P_RD_P0Q1	dddd dddd dddd dddd	0 (0x0000)
R0x360E	P_RD_P0Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3610	P_RD_P0Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3612	P_RD_P0Q4	dddd dddd dddd dddd	0 (0x0000)
R0x3614	P_BL_P0Q0	dddd dddd dddd dddd	0 (0x0000)
R0x3616	P_BL_P0Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3618	P_BL_P0Q2	dddd dddd dddd dddd	0 (0x0000)
R0x361A	P_BL_P0Q3	dddd dddd dddd dddd	0 (0x0000)
R0x361C	P_BL_P0Q4	dddd dddd dddd dddd	0 (0x0000)
R0x361E	P_GB_P0Q0	dddd dddd dddd dddd	0 (0x0000)
R0x3620	P_GB_P0Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3622	P_GB_P0Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3624	P_GB_P0Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3626	P_GB_P0Q4	dddd dddd dddd dddd	0 (0x0000)
R0x3640	P_GR_P1Q0	dddd dddd dddd dddd	0 (0x0000)

Table 3. MANUFACTURER SPECIFIC (continued)

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x3642	P_GR_P1Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3644	P_GR_P1Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3646	P_GR_P1Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3648	P_GR_P1Q4	dddd dddd dddd dddd	0 (0x0000)
R0x364A	P_RD_P1Q0	dddd dddd dddd dddd	0 (0x0000)
R0x364C	P_RD_P1Q1	dddd dddd dddd dddd	0 (0x0000)
R0x364E	P_RD_P1Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3650	P_RD_P1Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3652	P_RD_P1Q4	dddd dddd dddd dddd	0 (0x0000)
R0x3654	P_BL_P1Q0	dddd dddd dddd dddd	0 (0x0000)
R0x3656	P_BL_P1Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3658	P_BL_P1Q2	dddd dddd dddd dddd	0 (0x0000)
R0x365A	P_BL_P1Q3	dddd dddd dddd dddd	0 (0x0000)
R0x365C	P_BL_P1Q4	dddd dddd dddd dddd	0 (0x0000)
R0x365E	P_GB_P1Q0	dddd dddd dddd dddd	0 (0x0000)
R0x3660	P_GB_P1Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3662	P_GB_P1Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3664	P_GB_P1Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3666	P_GB_P1Q4	dddd dddd dddd dddd	0 (0x0000)
R0x3680	P_GR_P2Q0	dddd dddd dddd dddd	0 (0x0000)
R0x3682	P_GR_P2Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3684	P_GR_P2Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3686	P_GR_P2Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3688	P_GR_P2Q4	dddd dddd dddd dddd	0 (0x0000)
R0x368A	P_RD_P2Q0	dddd dddd dddd dddd	0 (0x0000)

Table 3. MANUFACTURER SPECIFIC (continued)

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x368C	P_RD_P2Q1	dddd dddd dddd dddd	0 (0x0000)
R0x368E	P_RD_P2Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3690	P_RD_P2Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3692	P_RD_P2Q4	dddd dddd dddd dddd	0 (0x0000)
R0x3694	P_BL_P2Q0	dddd dddd dddd dddd	0 (0x0000)
R0x3696	P_BL_P2Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3698	P_BL_P2Q2	dddd dddd dddd dddd	0 (0x0000)
R0x369A	P_BL_P2Q3	dddd dddd dddd dddd	0 (0x0000)
R0x369C	P_BL_P2Q4	dddd dddd dddd dddd	0 (0x0000)
R0x369E	P_GB_P2Q0	dddd dddd dddd dddd	0 (0x0000)
R0x36A0	P_GB_P2Q1	dddd dddd dddd dddd	0 (0x0000)
R0x36A2	P_GB_P2Q2	dddd dddd dddd dddd	0 (0x0000)
R0x36A4	P_GB_P2Q3	dddd dddd dddd dddd	0 (0x0000)
R0x36A6	P_GB_P2Q4	dddd dddd dddd dddd	0 (0x0000)
R0x36C0	P_GR_P3Q0	dddd dddd dddd dddd	0 (0x0000)
R0x36C2	P_GR_P3Q1	dddd dddd dddd dddd	0 (0x0000)
R0x36C4	P_GR_P3Q2	dddd dddd dddd dddd	0 (0x0000)
R0x36C6	P_GR_P3Q3	dddd dddd dddd dddd	0 (0x0000)
R0x36C8	P_GR_P3Q4	dddd dddd dddd dddd	0 (0x0000)
R0x36CA	P_RD_P3Q0	dddd dddd dddd dddd	0 (0x0000)
R0x36CC	P_RD_P3Q1	dddd dddd dddd dddd	0 (0x0000)
R0x36CE	P_RD_P3Q2	dddd dddd dddd dddd	0 (0x0000)
R0x36D0	P_RD_P3Q3	dddd dddd dddd dddd	0 (0x0000)
R0x36D2	P_RD_P3Q4	dddd dddd dddd dddd	0 (0x0000)
R0x36D4	P_BL_P3Q0	dddd dddd dddd dddd	0 (0x0000)

Table 3. MANUFACTURER SPECIFIC (continued)

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x36D6	P_BL_P3Q1	dddd dddd dddd dddd	0 (0x0000)
R0x36D8	P_BL_P3Q2	dddd dddd dddd dddd	0 (0x0000)
R0x36DA	P_BL_P3Q3	dddd dddd dddd dddd	0 (0x0000)
R0x36DC	P_BL_P3Q4	dddd dddd dddd dddd	0 (0x0000)
R0x36DE	P_GB_P3Q0	dddd dddd dddd dddd	0 (0x0000)
R0x36E0	P_GB_P3Q1	dddd dddd dddd dddd	0 (0x0000)
R0x36E2	P_GB_P3Q2	dddd dddd dddd dddd	0 (0x0000)
R0x36E4	P_GB_P3Q3	dddd dddd dddd dddd	0 (0x0000)
R0x36E6	P_GB_P3Q4	dddd dddd dddd dddd	0 (0x0000)
R0x3700	P_GR_P4Q0	dddd dddd dddd dddd	0 (0x0000)
R0x3702	P_GR_P4Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3704	P_GR_P4Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3706	P_GR_P4Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3708	P_GR_P4Q4	dddd dddd dddd dddd	0 (0x0000)
R0x370A	P_RD_P4Q0	dddd dddd dddd dddd	0 (0x0000)
R0x370C	P_RD_P4Q1	dddd dddd dddd dddd	0 (0x0000)
R0x370E	P_RD_P4Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3710	P_RD_P4Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3712	P_RD_P4Q4	dddd dddd dddd dddd	0 (0x0000)
R0x3714	P_BL_P4Q0	dddd dddd dddd dddd	0 (0x0000)
R0x3716	P_BL_P4Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3718	P_BL_P4Q2	dddd dddd dddd dddd	0 (0x0000)
R0x371A	P_BL_P4Q3	dddd dddd dddd dddd	0 (0x0000)
R0x371C	P_BL_P4Q4	dddd dddd dddd dddd	0 (0x0000)
R0x371E	P_GB_P4Q0	dddd dddd dddd dddd	0 (0x0000)

Table 3. MANUFACTURER SPECIFIC (continued)

(1 = read-only, always 1; 0 = read-only, always 0; d = programmable; ? = read-only, dynamic)

Register Hex	Name	Data Format (Binary)	Default Value Dec(Hex)
R0x3720	P_GB_P4Q1	dddd dddd dddd dddd	0 (0x0000)
R0x3722	P_GB_P4Q2	dddd dddd dddd dddd	0 (0x0000)
R0x3724	P_GB_P4Q3	dddd dddd dddd dddd	0 (0x0000)
R0x3726	P_GB_P4Q4	dddd dddd dddd dddd	0 (0x0000)
R0x3780	POLY_SC_ENABLE	d000 0000 0000 0000	0 (0x0000)
R0x3782	POLY_ORIGIN_C	0000 dddd dddd dddd	964 (0x03C4)
R0x3784	POLY_ORIGIN_R	0000 0ddd dddd dddd	604 (0x025C)
R0x3786	DIGITAL_CTRL_1	0000 00dd dddd dddd	6 (0x0006)
R0x37C0	P_GR_Q5	dddd dddd dddd dddd	0 (0x0000)
R0x37C2	P_RD_Q5	dddd dddd dddd dddd	0 (0x0000)
R0x37C4	P_BL_Q5	dddd dddd dddd dddd	0 (0x0000)
R0x37C6	P_GB_Q5	dddd dddd dddd dddd	0 (0x0000)

Attributes Columns; Usage and Values, left to right

Column 1; Buffering <blank> = Unbuffered S = Single Frame Synced D = Double Frame Synced	Column 2; Bad Frame <blank> = No bad frame Y = Causes a bad frame YM = Maskable bad frame	Column 3; Embedded <blank> = Not embedded E = embedded	Column 4; Locked <blank> = Not locked L = locked
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Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3000	15:0	0x0A56	CHIP_VERSION_REG (R/W)			E	
	Model ID. Read-only. Can be made read/write by clearing R0x301A-B[3].						
R0x3002	15:0	0x0008	Y_ADDR_START (R/W)	S	YM	E	
	The first row of visible pixels to be read out (not counting any dark rows that may be read). To move the image window, set this register to the starting Y value.						
R0x3004	15:0	0x0008	X_ADDR_START (R/W)	S		E	
	The first column of visible pixels to be read out (not counting any dark columns that may be read). To move the image window, set this register to the starting X value.						
R0x3006	15:0	0x04B7	Y_ADDR_END (R/W)	S	YM	E	
	The last row of visible pixels to be read out.						
R0x3008	15:0	0x0787	X_ADDR_END (R/W)	S		E	
	The last column of visible pixels to be read out.						
R0x300A	15:0	0x04C0	FRAME_LENGTH_LINES (R/W)	S	YM	E	
	The number of complete lines (rows) in the frame timing. This includes visible lines and vertical blanking lines.						
R0x300C	15:0	0x0264	LINE_LENGTH_PCK (R/W)	S	YM	E	
	The number of pixel clock periods in one line (row) time. This includes visible pixels and horizontal blanking time. Only even values are allowed.						
R0x300E	15:0	0x0000	REVISION_NUMBER (R/W)			E	
	Register to store the ROM/Chip Revision number. [7:0] = Part ID [11:8] = OTPM Revision [15:12] = CREV						
R0x3010	15:0	0xBEEF	LOCK_CONTROL (R/W)			E	
	This register protects the mirror mode select (register read mode). When set to value 0xBEEF, the horizontal and vertical mirror modes can be changed, otherwise these values are locked.						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3012	15:0	0x0010	COARSE_INTEGRATION_TIME (R/W)	S		E	
	Integration time specified in multiples of line_length_pck_.						
R0x3014	15:0	0x000A	FINE_INTEGRATION_TIME (R/W)	S		E	
	Fine integration time for context A. Fine integration is used to delay the shutter operation after the sample operation is finished. Thus, the integration time is increased. The resolution is 1 pixel clock time. Note that for short line length (R0x300C, R0x303E) values (1488 is shortest allowed), the available time for fine integration is limited. If programmed for more than available time, the normal sensor operation will be disrupted.						
R0x3016	15:0	0x0010	COARSE_INTEGRATION_TIME_CB (R/W)			E	
	Coarse integration time in context B.						
R0x3018	15:0	0x0000	FINE_INTEGRATION_TIME_CB (R/W)			E	
	Fine integration time in context B. Fine integration is used to delay the shutter operation after the sample operation is finished. Thus, the integration time is increased. The resolution is 1 pixel clock time. Note that for short line length (R0x300C, R0x303E) values (1488 is shortest allowed), the available time for fine integration is limited. If programmed for more than available time, the normal sensor operation will be disrupted.						
	15:0	0x2058	RESET_REGISTER (R/W)		Y	E	
	15	0x0001	GROUPED_PARAMETER_HOLD 0 = Update of many of the registers is synchronized to frame start. 1 = Inhibit register updates. Register changes will remain pending until this bit is returned to 0. When this bit is returned to 0, all pending register updates will be made on the next frame start.			E	
	14	X	Reserved				
	13	0x0001	Reserved				
	12	0x0000	SMIA_SERIALISER_DIS 0: Serial Interface Enabled. 1: Serial interface Disabled.			E	
	11	0x0000	FORCED_PLL_ON 0: PLL will be powered down when the sensor is in standby (low power mode). 1: PLL will be enabled even when the sensor is in "standby."			E	
	10	0x0000	RESTART_BAD 1: A restart is forced any time a bad frame is detected. This can shorten the delay when waiting for a good frame, since the delay for masking out a bad frame will be the integration time rather than the full-frame time.			E	
	9	0x0000	MASK_BAD 0: The sensor will produce bad (corrupted) frames as a result of some register changes. 1: Bad (corrupted) frames are masked within the sensor by extending the vertical blanking time for the duration of the bad frame.			E	

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x301A	8	0x0000	GPI_EN 0: The primary input buffers associated with the OUTPUT_ENABLE_N, TRIGGER input is powered down and cannot be used. 1: The input buffers are enabled and can be read through R0x3026-7.			E	
	7	0x0000	PARALLEL_EN 0: The parallel data interface (DOUT[11:0], LINE_VALID, FRAME_VALID, and PIXCLK) is disabled and the outputs are placed in a high-impedance state. 1: The parallel data interface is enabled. The output signals can be switched between a driven and a high-impedance state using output-enable control.			E	
	6	0x0001	DRIVE_PINS 0: The parallel data interface (DOUT[11:0], LINE_VALID, FRAME_VALID, and PIXCLK) may enter a high-impedance state (depending upon the enabling and use of the pad OUTPUT_ENABLE_N) 1: The parallel data interface is driven. This bit is "do not care" unless bit[7]=1.			E	
	5	0x0000	Reserved				
	4	0x0001	Reserved				
	3	0x0001	LOCK_REG Many parameter limitation registers that are specified as read-only are actually implemented as read/write registers. Clearing this bit allows such registers to be written.			E	
	2	0x0000	STREAM 0: Places the sensor in a low power mode. 1: Places the sensor in streaming mode. The result of clearing this bit depends upon the operating mode of the sensor. Entry and exit from streaming mode can also be controlled from the signal interface.	S		E	
	1	0x0000	RESTART This bit always reads as 0. Setting this bit causes the sensor to truncate the current frame at the end of the current row and start resetting (integrating) the first row. The delay before the first valid frame is read out is equal to the integration time.		Y	E	
	0	0x0000	RESET This bit always reads as 0. Setting this bit initiates a reset sequence: the frame being generated will be truncated.		Y	E	
	Controls the operation of the sensor. For details see the bit field descriptions.						
R0x301C	7:0	0x00	MODE_SELECT_ (R/W)		Y	E	
	7:1	X	Reserved				
	0	0x00	STREAM Setting this bit places the sensor in streaming mode. Clearing this bit places the sensor in a low power mode. The result of clearing this bit depends upon the operating mode of the sensor. Entry and exit from streaming mode can also be controlled from the signal interface.			E	
	Controls imaging modes of the sensor. For details see the bit field descriptions.						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x301D	7:0	0x00	IMAGE_ORIENTATION_ (R/W)	S	YM	E	
	7:2	X	Reserved				
	1	0x00	VERT_FLIP This bit is an alias of R0x3040[15].	S	YM	E	
	0	0x00	HORIZ_MIRROR This bit is an alias of R0x3040[14].	S	YM	E	
	Register to describe the readout of image array.						
R0x301E	15:0	0x002A	DATA_PEDESTAL (R/W)		Y	E	
	Constant offset that is added to pixel values at the end of datapath (after all corrections).						
R0x3021	7:0	0x00	SOFTWARE_RESET_ (R/W)		Y	E	
	This bit is an alias of R0x301A-B[0].						
R0x3022	7:0	0x00	GROUPED_PARAMETER_HOLD_ (R/W)		Y	E	
	Controls group parameters of the sensor. For details see the bit field descriptions.						
R0x3023	7:0	0x00	MASK_CORRUPTED_FRAMES_ (R/W)			E	
	7:1	X	Reserved				
	0	0x00	MASK_BAD 0 = The sensor will produce bad (corrupted) frames as a result of some register changes. 1 = Bad (corrupted) frames are masked within the sensor by extending the vertical blanking time for the duration of the bad frame.			E	
	When enabled, masks the first frame after any settings change which results in the first output frame being corrupted.						
R0x3024	7:0	0x00	PIXEL_ORDER_ (RO)			E	
	Status register describing the readout format of the pixel array						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3026	15:0	0x6500	GPI_STATUS (RO)			E	
	15:3	RO	Reserved				
	2	RO	TRIGGER Read-only. Returns the current state of the TRIGGER input pin. Invalid if R0x301A-B[8]=0.			E	
	1	RO	OUTPUT_ENABLE_N Read-only. Returns the current state of the OUTPUT_ENABLE_N input pin. Invalid if R0x301A-B[8]=0.			E	
	0	RO	SADDR Read-only. Returns the current state of the pin SADDR input pin. This pad is not controlled by gpi_en.			E	
	Reflects the status of the input pins: TRIGGER(2), OUTPUT_ENABLE_N(1), SADDR(0). Upper bits are hardwired to a constant.						
R0x3028	15:0	0x0010	ROW_SPEED (R/W)			E	
	Bits [6:4] of this register define the phase of the output pixclk. 2 sets of values are correct: a) 000, 010, 100, 110 => 0 delay (DOUT changes on rising edge of pixclk). b) 001, 011, 101, 111 => 1/2 clk delay (DOUT changes on falling edge of pixclk).						
R0x302A	15:0	0x0005	VT_PIX_CLK_DIV (R/W)			E	
	Sets the ratio of the serial output clock and sensor operation clock (P2 clock divider in PLL).						
R0x302C	15:0	0x0001	VT_SYS_CLK_DIV (R/W)			E	
	Sets the ratio of the VCO clk and the serial output clock (P1 divider in PLL).						
R0x302E	15:0	0x0003	PRE_PLL_CLK_DIV (R/W)			E	
	PLL input pre-divider value						
R0x3030	15:0	0x0032	PLL_MULTIPLIER (R/W)			E	
	PLL_MULTIPLIER						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3034	15:0	0x0000	CTX_CONTROL_REG (R/W)			E	
	15	0x0000	CONTEXT_LOAD Load context immediately			E	
	14:11	X	Reserved				
	10	0x0000	I2C_AUTO_INC_DISABLE Disable auto increment of addresses during i2c access			E	
	9	0x0000	CONTEXT_MULTI_CYCLE_MODE Enable multi frame context to cycle frame by frame between context 0 and context_multi_num.			E	
	8	0x0000	CONTEXT_MULTI_SYNC_MODE When set, multiple contexts are synced to the start of readout frames, else they are synced to the start of exposure frames			E	
	7	0x0000	CONTEXT_MULTI Enable Multiple context to be read out			E	
	6:4	0x0000	CONTEXT_MULTI_NUM Set number of consecutive contexts to read out when ctx_multi is set			E	
	3:0	0x0000	CONTEXT_SELECT Select context			E	
	Register Context Control Enabled to select from multiple contexts stored within the context RAM. Also supports ways to cycle through them on a frame by frame basis.						
R0x3036	15:0	0x000A	OP_PIX_CLK_DIV (R/W)		Y	E	
	Clock divisor applied to the output system clock to generate the output pixel clock.						
R0x3038	15:0	0x0001	OP_SYS_CLK_DIV (R/W)		Y	E	
	Clock divisor applied to PLL output clock to generate output system clock. Read-only.						
R0x303A	15:0	0xFFFF	FRAME_COUNT (RO)			E	
	Counts the number of output frames. At the startup is initialized to 0xFFFF.						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x303C	15:0	0x0000	FRAME_STATUS (RO)			E	
	15:4	X	Reserved				
	3	RO	PLL_LOCKED This bit indicates that the sensor is in streaming state and the PLL is locked. The sensor is now ready to output frames.			E	
	2	RO	FRAME_STATUS_FRAME_START_DURING_GPH Signifies that a new frame has started while group parameter hold is high. This tells the user if the sensor frame-sync start event has been missed, and therefore gains/integration time changes are going to take effect delayed by 1 frame.			E	
	1	RO	STANDBY_STATUS This bit indicates that the sensor is in standby state. It can be polled after standby is entered to see when the real low-power state is entered; which can happen at the end of row or frame depending on bit R0x301A[4].			E	
	0	RO	FRAMESYNC Set on register writes and reset on frame synchronization. Acts as debug flag to verify that register writes completed before last frame synchronization.			E	
	Read-only register describing the internal status of the sensor.						
R0x303E	15:0	0x0264	LINE_LENGTH_PCK_CB (R/W)	S		E	
	Line length in context B. The number of pixel clock periods in one line (row) time. This includes visible pixels and horizontal blanking time. Only even values are allowed. For smooth operation this would be the same value as LINE_LENGTH_PCK (0x300C).						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3040	15:0	0x0000	READ_MODE (R/W)	S	YM	E	
	15	0x0000	VERT_FLIP 0: Normal readout 1: Readout is flipped (mirrored) vertically so that the row specified by y_addr_end_ (+1) is read out of the sensor first. This register can only be changed when streaming is disabled	S	YM	E	
	14	0x0000	HORIZ_MIRROR 0: Normal readout 1: Readout is mirrored horizontally so that the column specified by x_addr_end_ (+1) is read out of the sensor first. This register can only be changed when streaming is disabled	S	YM	E	
	13	0x0000	READ_MODE_COL_BIN Column binning mode in context A. Pixel values are averaged in the digital domain. Use when skipping is enabled by setting x_odd_inc.	S		E	
	12	0x0000	READ_MODE_ROW_BIN Analog row binning control in context A. Use when row-wise skipping is enabled by setting y_odd_inc. The y_addr_start must be an even number when using row binning.	S		E	
	11	0x0000	READ_MODE_COL_BIN_CB Column binning mode in context B. Pixel values are averaged in the digital domain. Use when skipping is enabled by setting x_odd_inc.	S		E	
	10	0x0000	READ_MODE_ROW_BIN_CB Analog row binning control for context B. Use when row-wise skipping is enabled by setting y_odd_inc. The y_addr_start must be an even number when using row binning.	S		E	
	9	0x0000	READ_MODE_COL_SF_BIN_EN col_sf_bin_en Analog source follower column binning for context-A.	S		E	
	8	0x0000	READ_MODE_COL_SF_BIN_EN_CB col_sf_bin_en_cb Analog source follower column binning for context-B.	S		E	
	7	0x0000	READ_MODE_COL_SF_BIN_MONO_EN col_sf_bin_mono_en Analog source follower column binning with monochrome mode enabled for context-A.	S		E	
	6	0x0000	READ_MODE_COL_SF_BIN_MONO_EN_CB col_sf_bin_mono_en_cb Analog source follower column binning with monochrome mode enabled for context-B.	S		E	
	5	0x0000	READ_MODE_COL_SUM Column sum mode. Pixel values are summed in the digital domain. Use when skipping is enabled by setting x_odd_inc.	S		E	
	4:0	X	Reserved				
	RW register. Defines the readout mode from the pixel array.						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3042	15:0	0x0000	EXTRA_DELAY (R/W)	S		E	
	The last row in the frame is extended by the number of the sensor core clock periods specified here. The extra_delay must be configured to an even value. This register can be used to fine-tune the sensor maximum frame-rate.						
R0x3056	15:0	0x0080	GREEN1_GAIN (R/W)	S		E	
	Digital gain for Green1 (Gr) pixels in Context A, in format of xxxx.yyyyyyy.						
R0x3058	15:0	0x0080	BLUE_GAIN (R/W)	S		E	
	Digital gain for Blue pixels, in format of xxxx.yyyyyyy.						
R0x305A	15:0	0x0080	RED_GAIN (R/W)	S		E	
	Digital gain for Red pixels, in format of xxxx.yyyyyyy.						
R0x305C	15:0	0x0080	GREEN2_GAIN (R/W)	S		E	
	Digital gain for Green2 (Gb) pixels in Context A, in format of xxxx.yyyyyyy.						
R0x305E	15:0	0x0080	GLOBAL_GAIN (R/W)	S		E	
	Writing a gain to this register is equivalent to writing that code to each of the 4 color-specific gain registers. Reading from this register returns the value most recently written to the green1_gain register.						
R0x3060	15:0	0x000E	ANALOG_GAIN (R/W)	S		E	
	15	X	Reserved				
	14:12	0x0000	COARSE_GAIN_CB Coarse Analog gain in context B. Gain is 2 power of the register value.	S		E	
	11:8	0x0000	FINE_GAIN_CB Fine analog gain in context B	S		E	
	7	X	Reserved				
	6:4	0x0000	COARSE_GAIN Coarse Analog gain in context A. Gain is 2 power of the register value.	S		E	
	3:0	0x000E	FINE_GAIN Fine analog gain in context A.	S		E	
	Defines analog gains for both contexts						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3064	15:0	0x1802	SMIA_TEST (R/W)			E	
	15:13	X	Reserved				
	12	0x0001	Reserved				
	11:10	0x0002	Reserved				
	9	0x0000	Reserved				
	8	0x0000	EMBEDDED_DATA 0: Frames out of the sensor exclude the embedded data. 1: Frames of data out of the sensor include 2 rows of embedded data. This register field should only be change while the sensor is in software standby. Disabling the embedded data will not reduce the number of vertical blanking rows.			E	
	7	0x0000	EMBEDDED_STATS_EN 0: Embedded statistics are not transmitted on the 2 stats rows after the frame pixel data. 1: Embedded statistics are transmitted on the 2 stats data rows after the frame pixel data.			E	
	6:4	X	Reserved				
	3:0	0x0002	Reserved				
R0x3066	15:0	0x0000	CTX_WR_DATA_REG (R/W)			E	
	Context write data First control word: [15:8] = 1111_1000 (fixed) = 0xF8 [7:4] = Number of Contexts - this set the number of values (contexts) to store for each register address. A value of 0 indicates 1 context. You have two contexts, so this should be 0x1. [3:0] = upper address value. This set the value of A15:A12 of the register address that will be stored in the ram. For 0x30E8, this will be 0x3. So the value of the first control word should be 0xF813 (not 0xF823).						
R0x3068	15:0	0x0000	CTX_RD_DATA_REG (R/W)			E	
	context read data						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x306E	15:0	0x5C10	DATAPATH_SELECT (R/W)			E	
	15:13	0x0002	SLEW_RATE_CTRL_PARALLEL Selects the slew (edge) rate for the DOUT[11:0], FRAME_VALID, LINE_VALID and FLASH outputs. Only affects the FLASH output when parallel data output is disabled. The value (111) results in the fastest edge rates on these signals. Slowing down the edge rate can reduce ringing and electro-magnetic emissions.			E	
	12:10	0x0007	SLEW_RATE_CTRL_PIXCLK 3'b111 value is hardcoded in sensor for the fastest edge rate. So value change will have no effect.			E	
	9	0x0000	Reserved				
	8	0x0000	Reserved				
	7:5	X	Reserved				
	4	0x0000	Reserved				
	3	0x0000	Reserved				
	2	0x0000	Reserved				
	1:0	0x0000	SPECIAL_LINE_VALID 00: Normal behavior of LINE_VALID 01: LINE_VALID is driven continuously (continue generating LINE_VALID during vertical blanking) 10: LINE_VALID is driven continuously as LINE_VALID XOR FRAME_VALID 11: Reserved.			E	
R0x3070	15:0	0x0000	TEST_PATTERN_MODE (R/W)		Y	E	
			0: Normal operation. Generates output data from pixel array 1: Solid color test pattern. 2: Full color bar test pattern 3: Fade to gray color bar test pattern 256: Walking 1 test pattern (12-bit) Other: Reserved.				
R0x3072	15:0	0x0000	TEST_DATA_RED (R/W)		Y	E	
			The value for red pixels in the Bayer data used for the solid color test pattern and the test cursors.				
R0x3074	15:0	0x0000	TEST_DATA_GREENR (R/W)		Y	E	
			The value for green pixels in red/green rows of the Bayer data used for the solid color test pattern and the test cursors.				
R0x3076	15:0	0x0000	TEST_DATA_BLUE (R/W)		Y	E	
			The value for blue pixels in the Bayer data used for the solid color test pattern and the test cursors.				
R0x3078	15:0	0x0000	TEST_DATA_GREENB (R/W)		Y	E	
			The value for green pixels in blue/green rows of the Bayer data used for the solid color test pattern and the test cursors.				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3082	15:0	0x0003	OPERATION_MODE_CTRL (R/W)	S		E	
	15:2	X	Reserved				
	1:0	0x0003	OPERATION_MODE Operation Mode for Context A (Linear Mode Only) Bits[1:0]=3			E	
R0x3084	15:0	0x0003	OPERATION_MODE_CTRL_CB (R/W)	S		E	
	15:2	X	Reserved				
	1:0	0x0003	OPERATION_MODE_CB Operation Mode for Context B (Linear Mode Only) Bits[1:0]=3	S		E	
R0x3086	15:0	0x0000	SEQ_DATA_PORT (R/W)			E	
	Register used to write to or read from the sequencer RAM.						
R0x3088	15:0	0xC000	SEQ_CTRL_PORT (R/W)			E	
	15	RO	SEQUENCER_STOPPED Showing that sequencer is stopped (STANDBY mode) and the RAM is available for read or write.			E	
	14	0x0001	AUTO_INC_ON_READ 1: The access_address is incremented (by 1) after each read operation from seq_data_port (which returns only 1 byte).			E	
	13:10	X	Reserved				
	9:0	0x0000	ACCESS_ADDRESS When in STANDBY (not streaming) mode: address pointer to the sequencer RAM.			E	
	Register controlling the read and write to sequencer RAM.						
R0x308A	15:0	0x0008	X_ADDR_START_CB (R/W)			E	
	X_ADDR_START for context B						
R0x308C	15:0	0x0008	Y_ADDR_START_CB (R/W)			E	
	Y_ADDR_START for context B						
R0x308E	15:0	0x0787	X_ADDR_END_CB (R/W)			E	
	X_ADDR_END for context B						
R0x3090	15:0	0x04B7	Y_ADDR_END_CB (R/W)			E	
	Y_ADDR_END for context B						
R0x30A0	15:0	0x0001	X_EVEN_INC (RO)			E	
	Read-only.						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x30A2	15:0	0x0001	X_ODD_INC (R/W)	S	YM	E	
	1: No skip. 3: Skip 2. 7: Skip 4. 15: Skip 8. 31: Skip 16. Other values are not supported.						
R0x30A4	15:0	0x0001	Y_EVEN_INC (RO)			E	
	Read-only.						
R0x30A6	15:0	0x0001	Y_ODD_INC (R/W)	S	YM	E	
	1: No skip. 3: Skip 2. 7: Skip 4. 15: Skip 8. 31: Skip 16. Other values are not supported.						
R0x30A8	15:0	0x0001	Y_ODD_INC_CB (R/W)	S	YM	E	
	Y_ODD_INC context B						
R0x30AA	15:0	0x04C0	FRAME_LENGTH_LINES_CB (R/W)	S	YM	E	
	FRAME_LENGTH_LINES context B. See description for R0x300A						
R0x30AE	15:0	0x0003	X_ODD_INC_CB (R/W)	S	YM	E	
	X_ODD_INC context B						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x30B0	15:0	0x0028	DIGITAL_TEST (R/W)		Y	E	
	15	X	Reserved				
	14	0x0000	PLL_COMPLETE_BYPASS 0: PLL is enabled 1: PLL is bypassed. EXTCLK will be used. Note that the serial interface does not function when PLL is bypassed.			E	
	13	0x0000	CONTEXT_B Context Control. 0: Use Context A 1: Use Context B	S		E	
	12	0x0000	Reserved				
	11	0x0000	Reserved				
	10	0x0000	Reserved				
	9	0x0000	CONTEXT_SWITCH_OPERATION_MODE			E	
	8	0x0000	PIXCLK_ON When set, the parallel output PIXCLK will continue to toggle irrespective of stand-by mode.			E	
	7	0x0000	MONO_CHROME_OPERATION When set the CFA is monochrome and not color. Some features like skipping and corrections are affected. Useful in the bin2 mode.	S		E	
	6	0x0000	Reserved				
	5	0x0001	Reserved				
	4	0x0000	Reserved				
	3:2	0x0002	Reserved				
	1	0x0000	NO_SH_JUMP_LIMIT If set, integration time can be increased to any value on the fly without any maximum jump limit else, integration time would be increased in steps of maximum jump defined by $(FLL \cdot row_size \cdot \Delta dark_rows \cdot 4) \times 4$ though the integration time set is very high.			E	
	0	X	Reserved				
R0x30B2	15:0	0x0000	TEMPSENS_DATA_REG (R/W)	S		E	
	Output value from temperature sensor.						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x30B4	15:0	0x0000	TEMPSENS_CTRL_REG (R/W)			E	
	15:6	0x0000	RETRIGGER_THRESHOLD When the measured absolute temperature (ADC value) changes more than this setting, the delta dark algorithm is retrIGGERED and the temperature is saved as the comparison level for the next measurement. If the value is set to zero, the retrigger function will be disabled.			E	
	5	0x0000	TEMP_CLEAR_VALUE Clear data register (sanity check).			E	
	4	0x0000	TEMP_START_CONVERSION When asserted, a new temp value will be generated for each frame capture. When asserted in standby mode, a new temp value will be generated.			E	
	3:1	0x0000	Reserved				
	0	0x0000	TEMPSENS_POWER_ON 0: Temperature sensor power off 1: Temperature sensor power on			E	
			Control register for temperature sensor				
R0x30BC	15:0	0x0080	GREEN1_GAIN_CB (R/W)	S		E	
			Digital gain for Green1 (Gr) pixels in Context B, in format of xxxx.yyyyyyy.				
R0x30BE	15:0	0x0080	BLUE_GAIN_CB (R/W)	S		E	
			Digital gain for Blue pixels in Context B, in format of xxxx.yyyyyyy.				
R0x30C0	15:0	0x0080	RED_GAIN_CB (R/W)	S		E	
			Digital gain for Red pixels in Context B, in format of xxxx.yyyyyyy.				
R0x30C2	15:0	0x0080	GREEN2_GAIN_CB (R/W)	S		E	
			Digital gain for Green2 (Gb) pixels in Context B, in format of xxxx.yyyyyyy.				
R0x30C4	15:0	0x0080	GLOBAL_GAIN_CB (R/W)	S		E	
			Writing a gain to this register is equivalent to writing that code to each of the 4 color-specific gain registers in Context B. Reading from this register returns the value most recently written to the green1_gain register.				
R0x30C6	15:0	0x0000	TEMPSENS_CALIB1 (R/W)			E	
			Temperature sensor calibration register 1				
R0x30C8	15:0	0x0000	TEMPSENS_CALIB2 (R/W)			E	
			Temperature sensor calibration register 2				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x30CE	15:0	0x0000	GRR_CONTROL1 (R/W)			E	
	15:10	X	Reserved				
	9	0x0000	Reserved				
	8	0x0000	SLAVE_SH_SYNC_MODE Also called SURROUND VIEW mode. When set and the sensor is set to stream mode, it waits for an external trigger signal. When a trigger is received, a time interval of frame length lines is started. The shutter pointers are started at appropriate point and when this time interval is finished, the read out is started. The external trigger period must not be less than frame length lines, otherwise the triggers happening during the time interval will be ignored.			E	
	7	0x0000	Reserved				
	6	0x0000	Reserved				
	5	0x0000	FRAME_START_MODE This bit controls the frame timing of the first frame when entering stream mode (or trigger mode). When low, the integration time is started and then readout begins. It means that the first frame will start after a delay equal to CIT. When high, a frame time is started and at the appropriate time point, the integration starts. It means that the first frame will start after one frame time.			E	
	4	0x0000	Reserved				
	3	X	Reserved				
	2	0x0000	Reserved				
	1	X	Reserved				
	0	0x0000	Reserved				
R0x30FE	15:0	0x002A	NOISE_PEDESTAL (R/W)			E	
	Pedestal added prior to memory operations. Noise pedestal and AdaCD pedestals should be set to the same value.						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3100	15:0	0x0000	AECTRLREG (R/W)	S		E	
	15:8	X	Reserved				
	7	0x0000	Reserved				
	6:5	0x0000	MIN_ANA_GAIN Minimum analogue gain to be used by AE. '00'=1x (default) '01'=2x '10'=4x '11'=8x			E	
	4	0x0000	AUTO_DG_EN Automatic control of digital gain by AE is enabled. Automatic digital gain control should be turned off (auto_dg_en = 0) all the times for AR0234, preventing the Auto Exposure algorithm from adjusting the digital gain.			E	
	3	X	Reserved				
	2	0x0000	Reserved				
	1	0x0000	AUTO_AG_EN When set, enables the automatic ae control of analogue gain.			E	
	0	0x0000	AE_ENABLE 1 => enables the on-chip AE algorithm			E	
R0x3102	15:0	0x0500	AE_LUMA_TARGET_REG (R/W)			E	
	Average Gr target value to be reached by the auto exposure						
R0x3108	15:0	0x0070	AE_MIN_EV_STEP_REG (R/W)			E	
	Minimum exposure value step size [15:8] : Reserved [7:0] : Min_EV_stepsize = (min step size)*256. Since Min_EV_step sizes are small and they are typically less than 1 e.g. 1/16, 7/16 ... these are multiplied by 256 and then the value is written to this register.						
R0x310A	15:0	0x0008	AE_MAX_EV_STEP_REG (R/W)			E	
	Maximum exposure value step size. Note that since this value is always greater than 1 there is no need to multiply by 256 as in the case of min_EV_stepsize.						
R0x310C	15:0	0x0200	AE_DAMP_OFFSET_REG (R/W)			E	
	Adjusts step size and settling speed.						
R0x310E	15:0	0x2000	AE_DAMP_GAIN_REG (R/W)			E	
	Adjusts step size and settling speed.						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3110	15:0	0x0140	AE_DAMP_MAX_REG (R/W)			E	
			Max value allowed for recursiveDamp (multiplied by 256 since internal value is typically <1). For most applications, the value of recursiveDamp should be <1, otherwise AE will overshoot the target. For applications with fast settling required, it may be desirable to allow recursiveDamp >1. Default value: $0.875 * 256 = 0x00E0$				
R0x311C	15:0	0x02A0	AE_MAX_EXPOSURE_REG (R/W)			E	
			Maximum integration (exposure) time in rows to be used by AE.				
R0x311E	15:0	0x0001	AE_MIN_EXPOSURE_REG (R/W)			E	
			Minimum integration (exposure) time in rows to be used by AE.				
R0x3124	15:0	0x7FFF	AE_DARK_CUR_THRESH_REG (R/W)			E	
			The dark current level that stops AE from increasing integration time. Note that increased integration time would increase dark current as well and signal level (SNR) would drop because photo diode well capacity is limited.				
R0x312A	15:0	0x0080	AE_CURRENT_GAINS (RO)			E	
	15:14	X	Reserved				
	13:11	RO	AE_ANA_GAIN The gain decided by AE, when it is enabled and can control the analogue gain.			E	
	10:0	RO	AE_DIG_GAIN The gain decided by AE, when it is enabled and can control the digital gain.			E	
			Shows the gain settings decided by AE.				
R0x312E	15:0	0x0000	ACTUAL_FRAME_LENGTH_LINES (RO)			E	
R0x3140	15:0	0x0000	AE_ROI_X_START_OFFSET (R/W)			E	
			Number of pixels into each row before the ROI starts NOTE: if statistics are being gathered from a scaled image then 'number of pixels' value must be the number of scaled pixels				
R0x3142	15:0	0x0000	AE_ROI_Y_START_OFFSET (R/W)			E	
			Number of rows into each frame before the ROI starts				
R0x3144	15:0	0x0780	AE_ROI_X_SIZE (R/W)			E	
			Number of columns in the ROI				
R0x3146	15:0	0x04B0	AE_ROI_Y_SIZE (R/W)			E	
			Number of rows in the ROI				
R0x3148	15:0	0x03D0	AE_HIST_BEGIN_PERC (R/W)			E	
			Defines the percentage of Gr pixels that must have values below hist_begin. Specified as a number < 1 = 0.xx...xx				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x314A	15:0	0xFFFF	AE_HIST_END_PERC (R/W)			E	
			Defines the percentage of Gr pixels that must have values below hist_end. Specified as a number < 1 = 0.xx...xx. A value of all 1s is treated as a special case and equates to 1.0 (100%)				
R0x314C	15:0	0x0100	AE_HIST_DIV (R/W)			E	
			Defines the point at which the histogram is divided into the low and high end. Boundary value = hist_div*16				
R0x314E	15:0	0x0020	AE_NORM_WIDTH_MIN (R/W)			E	
			Defines the minimum histogram width normalization factor (=norm_width_min*16), for norm_abs_dev calculation. A value of all 1s turns off the norm_width_min option ie. all absolute deviation is normalized by hist_end - hist_begin				
R0x3150	15:0	0x0000	AE_MEAN_H (RO)			E	
			The true mean of all Gr pixels in the ROI (higher bits)				
R0x3152	15:0	0x0000	AE_MEAN_L (RO)			E	
			The true mean of all Gr pixels in the ROI (16 least significant bits)				
R0x3154	15:0	0x0000	AE_HIST_BEGIN_H (RO)			E	
			Code value corresponding to the histogram bin below which(hist_begin_perc*100)% of pixels exist (higher bits)				
R0x3156	15:0	0x0000	AE_HIST_BEGIN_L (RO)			E	
			Code value corresponding to the histogram bin below which (hist_begin_perc*100)% of pixels exist (lower 16 bits)				
R0x3158	15:0	0x0000	AE_HIST_END_H (RO)			E	
			Code value corresponding to the histogram bin below which(hist_end_perc*100)% of pixels exist (higher bits)				
R0x315A	15:0	0x0000	AE_HIST_END_L (RO)			E	
			Code value corresponding to the histogram bin below which(hist_end_perc*100)% of pixels exist (lower 16 bits)				
R0x315C	15:0	0x0000	AE_LOW_END_MEAN_H (RO)			E	
			The true mean of all Gr pixels in the ROI that fall into the low end of the histogram (where low end is defined by hist_div) (higher bits)				
R0x315E	15:0	0x0000	AE_LOW_END_MEAN_L (RO)			E	
			The true mean of all Gr pixels in the ROI that fall into the low end of the histogram (where low end is defined by hist_div) (lower 16 bits)				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3160	15:0	0x0000	AE_PERC_LOW_END (RO)			E	
			Percentage of Gr pixels in ROI that fall into the low end of the histogram. Specified as a number $< 1 = 0.xx...xxx$				
R0x3162	15:0	0x0000	AE_NORM_ABS_DEV (RO)			E	
			Percentage of Gr pixels in ROI that fall into the low end of the histogram. Specified as a number $< 1 = 0.xx...xxx$				
R0x3164	15:0	0x0001	AE_COARSE_INTEGRATION_TIME (RO)			E	
			Reads the current frame integration time in rows.				
R0x3166	15:0	0x03DA	AE_AG_EXPOSURE_HI (R/W)			E	
			At this integration time, the analog gain is increased (when AE is enabled to control also the analog gain).				
R0x3168	15:0	0x01A3	AE_AG_EXPOSURE_LO (R/W)			E	
			At this integration time, the AE is reduced (when AE is enabled to control the analog gain also),				
R0x3180	15:0	0x824F	DELTA_DK_CONTROL (R/W)			E	
	15	0x0001	DELTA_DK_SUB_EN Enables the delta dark correction.			E	
	14	0x0000	DELTA_DK_EVERY_FRAME Running the delta dark algorithm every frame or when gain, integration time is changing.			E	
	13	0x0000	DELTA_DK_RECALC Forces recalculation of the delta dark value.			E	
	12	0x0000	Reserved				
	11	0x0000	Reserved				
	10	0x0000	Reserved				
	9	0x0001	Reserved				
	8	X	Reserved				
	7:4	0x0004	Reserved				
	3:0	0x000F	Reserved				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x31AC	15:0	0x0A0A	DATA_FORMAT_BITS (R/W)	S			
	15:13	X	Reserved				
	12:8	0x000A	DATA_FORMAT_IN The bit-width of the pixel data pre compression or truncation				
	7:5	X	Reserved				
	4:0	0x000A	DATA_FORMAT_OUT The bit-width of the pixel data post compression or truncation. 0x08: 8-bit 0x0A: 10-bit				
	Data formats pre and post compression or truncation. Compression is selected in COMPANDING 0x31D0. Truncation will be selected by setting register value to 0x0A08						
R0x31AE	15:0	0x0204	SERIAL_FORMAT (R/W)				
	The upper byte of this register (interface type) is read-only. When the serial interface is enabled (reset_register[12]=0), the lower byte configures the number of Serial lanes to be used: 4: Quad-lane MIPI 2: Dual-lane MIPI 1: Single-lane MIPI						
R0x31B0	15:0	0x0070	FRAME_PREAMBLE (R/W)				
	Frame preamble This timing value, expressed in op_pix_clk periods, must be large enough to allow the MIPI wakeup and start-of-frame short packet to be transmitted prior to the start of a frame of pixel data. The default value should be correct for most applications. Too small a value will result in an INSUFFICIENT_FRAME_PREAMBLE error being flagged in the DATAPATH_STATUS register. Legal values: [0, 255].						
R0x31B2	15:0	0x0051	LINE_PREAMBLE (R/W)				
	Line preamble This timing value, expressed in op_pix_clk periods, must be large enough to allow the MIPI long packet header to be transmitted prior to the start of a line of pixel data. The default value should be correct for most applications. Too small a value will result in an INSUFFICIENT_LINE_PREAMBLE error being flagged in the DATAPATH_STATUS register. Legal values: [0, 255].						
R0x31B4	15:0	0x5248	MIPI_TIMING_0 (R/W)				
	15:12	0x0005	T_HS_PREPARE t_hs_prepare				
	11	0x0000	Reserved				
	10:6	0x0009	T_HS_TRAIL Time, in op_pix_clk				
	5	0x0000	Reserved				
	4:0	0x0008	T_CLK_TRAIL Time, in op_pix_clk				
	MIPI timing parameters						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x31B6	15:0	0x3298	MIPI_TIMING_1 (R/W)				
	15:12	0x0003	T_CLK_PREPARE Time, in op_pix_clk periods, to drive LP-00 prior to entering HS data transmission mode.				
	11:6	0x000A	T_HS_EXIT Time, in op_pix_clk periods, to drive LP-11 after HS burst.				
	5:0	0x0018	T_HS_ZERO Time, in op_pix_clk periods, to drive HS-0 before the sync sequence.				
	MIPI timing parameters						
R0x31B8	15:0	0x904B	MIPI_TIMING_2 (R/W)				
	15:12	0x0009	T_BGAP t_bgap lower 4 bits. Bandgap settling time				
	11:6	0x0001	T_CLK_PRE Time, in op_pix_clk periods, of any low-power state period.				
	5:0	0x000B	T_CLK_POST Time, in op_pix_clk periods, to drive the HS clock after the data lane has gone into low-power mode.				
	MIPI timing parameters						
R0x31BA	15:0	0x030C	MIPI_TIMING_3 (R/W)				
	15:14	0x0000	Reserved				
	13	X	MIPI_TIMING_3_RESERVED Reserved				
	12:7	0x0006	T_LPX Time, in op_pix_clk periods, of any low-power state period				
	6:0	0x000C	T_WAKE_UP Time to recover from ultra low-power mode (ULPM). ULPM is exited by applying a mark state for (8192) * T_WAKE_UP * op_pix_clk.				
	MIPI timing parameters						
R0x31BC	15:0	0x8E8A	MIPI_TIMING_4 (R/W)				
	15	0x0001	CONT_TX_CLK Set continuous clock mode.				
	14	0x0000	HEAVY_LP_LOAD heavy_lp_load				
	13:7	0x001D	Reserved				
	6:0	0x000A	T_INIT Initialization time when first entering stop state (LP-11) after power-up or reset. LP-11 is transmitted for a minimum of (1024) * T_INIT * op_pix_clk.				
	MIPI timing parameters						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x31BE	15:0	0x0023	SERIAL_CONFIG_STATUS (R/W)				
	15:10	X	Reserved				
	9	0x0000	Reserved				
	8:4	0x0002	Reserved				
	3:2	0x0000	Reserved				
	1	0x0001	FRAME_CNT_RST Resets the MIPI frame counting function				
	0	0x0001	FRAME_CNT_EN Enables MIPI frame counting function				
	Serial configuration and status						
R0x31C6	15:0	0x8000	SERIAL_CONTROL_STATUS (R/W)				
	15	RO	Reserved				
	14	RO	Reserved				
	13	0x0000	Reserved				
	12	X	Reserved				
	11:10	0x0000	Reserved				
	9	0x0000	Reserved				
	8	0x0000	Reserved				
	7	0x0000	Reserved				
	6:4	X	Reserved				
	3	0x0000	Reserved				
	2	0x0000	Reserved				
	1	0x0000	Reserved				
	0	0x0000	VERT_LEFT_BAR_EN An optional filler code of 1s may be padded after the sync code. When filler codes are enabled, the receiver must window the received image to eliminate the first data word per data lane columns per PHYs).				
	See descriptions in the bit fields.						
R0x31C8	15:0	0xFFFF	SERIAL_CRC_0 (RO)			E	
	Used to capture the full frame CRC						
R0x31D0	15:0	0x0000	COMPANDING (R/W)			E	
	15:4	X	Reserved				
	3	0x0000	Reserved				
	2:1	X	Reserved				
	0	0x0000	DPCM_EN Enables DPCM compression if DATA_FORMAT_BITS 0x31AC is set to 0x0A08 .			E	

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x31D2	15:0	0x0000	STAT_FRAME_ID (R/W)			E	
			User programmable Stats Frame ID, Transferred in the first line of stats along with frame-count				
R0x31D6	15:0	0xFFFF	I2C_WRT_CHECKSUM (R/W)			E	
			Checksum of I2C write operations.				
R0x31D8	15:0	0x0000	SERIAL_TEST (R/W)			E	
	15:12	X	Reserved				
	11:8	0x0000	TEST_LANE_EN Defines which data lanes are enabled when test_en = 1 - b0 = data lane 0 ... b3 = data lane 3			E	
	7:3	0x0000	TEST_MODE Define test mode to be applied to MIPI/CCP interface if test_en is asserted: MIPI: 0 = Transmit LP-00 on all enabled data and clock lanes 1 = Transmit LP-11 on all enabled data and clock lanes 2 = Transmit HS-0 on all enabled data and clock lanes 3 = Transmit HS-1 on all enabled data and clock lanes 4 = Transmit a square wave at half the potential serial data rate on all enabled data and clock lanes 5 = Transmit a square wave at the pixel data rate on all enabled data and clock lanes 6 = Transmit a LP square wave at half the pixel data rate on all enabled data and clock lanes 7 = Transmit a continuous , repeated , sequence of prbs31 (non-packetized) copied on all enabled data lanes 8 = Transmit a continuous , repeated , sequence of prbs9 data (non-packetized) copied on all enabled data lanes			E	
	2:0	0x0000	ALTERNATE_TEST_MODE Alternate column test mode x x 0 : _disable_test x 0 1 : Enable test - even column select even frame - even column , odd frame - odd column x 1 1 : Enable test - odd column select even frame - odd column , odd frame - Even column 1 0 1 : Enable test in sticky mode - even column/ frame 1 1 1 : Enable test in sticky mode - odd column/ frame			E	

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x31E0	15:0	0x0000	PIX_DEF_ID (R/W)			E	
	15	0x0000	Reserved				
	14	0x0000	Reserved				
	13:4	X	Reserved				
	3	0x0000	PIX_DEF_1D_DDC_EN If set, enables Single Pixel Dynamic Defect Correction (1D-DDC).		Y	E	
	2	0x0000	PIX_DEF_1D_DDC_EDGE_DETECT_EN If set, enables Edge Detection for 1D-DDC.		Y	E	
	1	0x0000	CORRECTION_MODE Mode of pixel defect correction. 0: Tags bad pixels with the reserved value 0. 1: Corrects bad pixels		Y	E	
	0	0x0000	ENABLE Enable pixel defect correction.		Y	E	
R0x31E8	15:0	0x0000	HORIZONTAL_CURSOR_POSITION (R/W)			E	
	Specifies the start row for the test cursor.						
R0x31EA	15:0	0x0000	VERTICAL_CURSOR_POSITION (R/W)			E	
	Specifies the start column for the test cursor.						
R0x31EC	15:0	0x0000	HORIZONTAL_CURSOR_WIDTH (R/W)			E	
	Specifies the width, in rows, of the horizontal test cursor. A width of 0 disables the cursor.						
R0x31EE	15:0	0x0000	VERTICAL_CURSOR_WIDTH (R/W)			E	
	Specifies the width, in columns, of the vertical test cursor. A width of 0 disables the cursor.						
R0x31F0	15:0	0x0000	AE_STATS_CONTROL (R/W)			E	
	15:4	X	Reserved				
	3	0x0000	AE_COMPAND_EN If set, stats companding is enabled and 18-bit stats output is companded to 16-bit.			E	
	2	0x0000	Reserved				
	1:0	0x0000	COLOR_SEL Select one color in the Bayer pattern for which the statistics is collected. 0: GreenRed 1: Red 2: Blue 3: GreenBlue			E	
R0x31FC	15:0	0x3020	CCI_IDS (R/W)			E	
	CCI addresses.						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x31FE	15:0	0x0923	CUSTOMER_REV (RO)			E	
			Customer revision [3:0] = CREV [6:4] = CFA [9:7] = CRA [13:10] = Minor Rev [15:14] = Open				
R0x3240	15:0	0x0100	AE_X1_START_OFFSET (R/W)				
			This register is used to define the grid for stats computation. It sets the start column address for the second grid in x- direction. Note This register value must be a multiple of four.				
R0x3242	15:0	0x0200	AE_X2_START_OFFSET (R/W)				
			This register is used to define the grid for stats computation. It sets the start column address for the third grid in x- direction. Note This register value must be a multiple of four.				
R0x3244	15:0	0x0300	AE_X3_START_OFFSET (R/W)				
			This register is used to define the grid for stats computation. It sets the start column address for the fourth grid in x- direction. Note This register value must be a multiple of four.				
R0x3246	15:0	0x0400	AE_X4_START_OFFSET (R/W)				
			This register is used to define the grid for stats computation. It sets the start column address for the fifth grid in x- direction. Note This register value must be a multiple of four.				
R0x3248	15:0	0x00A0	AE_Y1_START_OFFSET (R/W)				
			This register is used to define the grid for stats computation. It sets the start row address for the second grid in y- direction.				
R0x324A	15:0	0x0140	AE_Y2_START_OFFSET (R/W)				
			This register is used to define the grid for stats computation. It sets the start row address for the third grid in y- direction.				
R0x324C	15:0	0x01E0	AE_Y3_START_OFFSET (R/W)				
			This register is used to define the grid for stats computation. It sets the start row address for the fourth grid in y- direction.				
R0x324E	15:0	0x0280	AE_Y4_START_OFFSET (R/W)				
			This register is used to define the grid for stats computation. It sets the start row address for the fifth grid in y- direction.				
R0x3250	15:0	0xFFFF	AE_GRID_SEL_LO (R/W)				
			This register is used to select first 16 grids (out of total 25 grids) for the stats calculation. If any of the bit is set to 1 then the corresponding grid is included for the stats calculation				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3252	15:0	0x01FF	AE_GRID_SEL_HI (R/W)				
	This register is used to select last 9 grids (out of total 25 grids) for the stats calculation. If any of the bit is set to 1 then the corresponding grid is included for the stats calculation						
R0x3270	15:0	0x0000	LED_FLASH_CONTROL (R/W)				
	15:9	X	Reserved				
	8	0x0000	LED_FLASH_EN Enables LED Flash when set				
	7:0	0x0000	LED_DELAY LED lead/lag delay in terms of (row_time/2). It is a two's compliment number If Bit[7] is 0, LED Flash will be lagging the Integration Start Point If Bit[7] is 1, LED Flash will be leading the Integration Start Point				
R0x3338	15:0	0x0000	MIPI_TEST_CNTRL (R/W)				
	Bits to test CMOS in and out						
R0x333A	15:0	0x3231	MIPI_COMPRESS_8_DATA_TYPE (R/W)				
	mipi_compress_8_data_type						
R0x333C	15:0	0x3534	MIPI_COMPRESS_7_DATA_TYPE (R/W)				
	mipi_compress_7_data_type						
R0x333E	15:0	0x3736	MIPI_COMPRESS_6_DATA_TYPE (R/W)				
	mipi_compress_6_data_type						
R0x3340	15:0	0x3330	MIPI_JPEG_PN9_DATA_TYPE (R/W)				
	mipi_jpeg_pn9_data_type						
R0x3354	15:0	0x002B	MIPI_CNTRL (R/W)				
	15:9	X	Reserved				
	8	0x0000	Reserved				
	7:6	0x0000	CHAN_NUM Channel number to be inserted in the MIPI packet header				
	5:0	0x002B	DATA_TYPE_XMIT Data type Encoded as per MIPI CSI-2 spec for 10 bit -> 0x2B for 8 bit -> 0x2A				
	mipi_cntrl						
R0x3356	15:0	0x0000	MIPI_TEST_PATTERN_CNTRL (R/W)				
	Data to be output on pad DN1, DP1, DN0, DP0, CN and CP when configured as a CMOS output for MIPI						

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3358	15:0	0x003F	MIPI_TEST_PATTERN_STATUS (RO)				
			Data input via cmos input pad DN1, DP1, DN0, DP0, CN and CP				
R0x3600	15:0	0x0000	P_GR_P0Q0 (R/W)				
			P0 coefficient for Q0 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3602	15:0	0x0000	P_GR_P0Q1 (R/W)				
			P0 coefficient for Q1 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3604	15:0	0x0000	P_GR_P0Q2 (R/W)				
			P0 coefficient for Q2 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3606	15:0	0x0000	P_GR_P0Q3 (R/W)				
			P0 coefficient for Q3 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3608	15:0	0x0000	P_GR_P0Q4 (R/W)				
			P0 coefficient for Q4 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x360A	15:0	0x0000	P_RD_P0Q0 (R/W)				
			P0 coefficient for Q0 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x360C	15:0	0x0000	P_RD_P0Q1 (R/W)				
			P0 coefficient for Q1 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x360E	15:0	0x0000	P_RD_P0Q2 (R/W)				
			P0 coefficient for Q2 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3610	15:0	0x0000	P_RD_P0Q3 (R/W)				
			P0 coefficient for Q3 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3612	15:0	0x0000	P_RD_P0Q4 (R/W)				
			P0 coefficient for Q4 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3614	15:0	0x0000	P_BL_P0Q0 (R/W)				
			P0 coefficient for Q0 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x3616	15:0	0x0000	P_BL_P0Q1 (R/W)				
			P0 coefficient for Q1 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x3618	15:0	0x0000	P_BL_P0Q2 (R/W)				
			P0 coefficient for Q2 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x361A	15:0	0x0000	P_BL_P0Q3 (R/W)				
			P0 coefficient for Q3 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x361C	15:0	0x0000	P_BL_P0Q4 (R/W)				
			P0 coefficient for Q4 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x361E	15:0	0x0000	P_GB_P0Q0 (R/W)				
			P0 coefficient for Q0 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3620	15:0	0x0000	P_GB_P0Q1 (R/W)				
			P0 coefficient for Q1 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3622	15:0	0x0000	P_GB_P0Q2 (R/W)				
			P0 coefficient for Q2 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3624	15:0	0x0000	P_GB_P0Q3 (R/W)				
			P0 coefficient for Q3 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3626	15:0	0x0000	P_GB_P0Q4 (R/W)				
			P0 coefficient for Q4 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3640	15:0	0x0000	P_GR_P1Q0 (R/W)				
			P1 coefficient for Q0 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3642	15:0	0x0000	P_GR_P1Q1 (R/W)				
			P1 coefficient for Q1 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3644	15:0	0x0000	P_GR_P1Q2 (R/W)				
			P1 coefficient for Q2 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3646	15:0	0x0000	P_GR_P1Q3 (R/W)				
			P1 coefficient for Q3 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3648	15:0	0x0000	P_GR_P1Q4 (R/W)				
			P1 coefficient for Q4 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x364A	15:0	0x0000	P_RD_P1Q0 (R/W)				
			P1 coefficient for Q0 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x364C	15:0	0x0000	P_RD_P1Q1 (R/W)				
			P1 coefficient for Q1 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x364E	15:0	0x0000	P_RD_P1Q2 (R/W)				
			P1 coefficient for Q2 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3650	15:0	0x0000	P_RD_P1Q3 (R/W)				
			P1 coefficient for Q3 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3652	15:0	0x0000	P_RD_P1Q4 (R/W)				
			P1 coefficient for Q4 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3654	15:0	0x0000	P_BL_P1Q0 (R/W)				
			P1 coefficient for Q0 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x3656	15:0	0x0000	P_BL_P1Q1 (R/W)				
			P1 coefficient for Q1 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3658	15:0	0x0000	P_BL_P1Q2 (R/W)				
			P1 coefficient for Q2 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x365A	15:0	0x0000	P_BL_P1Q3 (R/W)				
			P1 coefficient for Q3 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x365C	15:0	0x0000	P_BL_P1Q4 (R/W)				
			P1 coefficient for Q4 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x365E	15:0	0x0000	P_GB_P1Q0 (R/W)				
			P1 coefficient for Q0 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3660	15:0	0x0000	P_GB_P1Q1 (R/W)				
			P1 coefficient for Q1 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3662	15:0	0x0000	P_GB_P1Q2 (R/W)				
			P1 coefficient for Q2 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3664	15:0	0x0000	P_GB_P1Q3 (R/W)				
			P1 coefficient for Q3 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3666	15:0	0x0000	P_GB_P1Q4 (R/W)				
			P1 coefficient for Q4 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3680	15:0	0x0000	P_GR_P2Q0 (R/W)				
			P2 coefficient for Q0 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3682	15:0	0x0000	P_GR_P2Q1 (R/W)				
			P2 coefficient for Q1 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3684	15:0	0x0000	P_GR_P2Q2 (R/W)				
			P2 coefficient for Q2 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3686	15:0	0x0000	P_GR_P2Q3 (R/W)				
			P2 coefficient for Q3 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3688	15:0	0x0000	P_GR_P2Q4 (R/W)				
			P2 coefficient for Q4 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x368A	15:0	0x0000	P_RD_P2Q0 (R/W)				
			P2 coefficient for Q0 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x368C	15:0	0x0000	P_RD_P2Q1 (R/W)				
			P2 coefficient for Q1 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x368E	15:0	0x0000	P_RD_P2Q2 (R/W)				
			P2 coefficient for Q2 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3690	15:0	0x0000	P_RD_P2Q3 (R/W)				
			P2 coefficient for Q3 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3692	15:0	0x0000	P_RD_P2Q4 (R/W)				
			P2 coefficient for Q4 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3694	15:0	0x0000	P_BL_P2Q0 (R/W)				
			P2 coefficient for Q0 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x3696	15:0	0x0000	P_BL_P2Q1 (R/W)				
			P2 coefficient for Q1 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x3698	15:0	0x0000	P_BL_P2Q2 (R/W)				
			P2 coefficient for Q2 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x369A	15:0	0x0000	P_BL_P2Q3 (R/W)				
			P2 coefficient for Q3 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x369C	15:0	0x0000	P_BL_P2Q4 (R/W)				
			P2 coefficient for Q4 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x369E	15:0	0x0000	P_GB_P2Q0 (R/W)				
			P2 coefficient for Q0 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x36A0	15:0	0x0000	P_GB_P2Q1 (R/W)				
			P2 coefficient for Q1 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x36A2	15:0	0x0000	P_GB_P2Q2 (R/W)				
			P2 coefficient for Q2 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x36A4	15:0	0x0000	P_GB_P2Q3 (R/W)				
			P2 coefficient for Q3 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x36A6	15:0	0x0000	P_GB_P2Q4 (R/W)				
			P2 coefficient for Q4 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x36C0	15:0	0x0000	P_GR_P3Q0 (R/W)				
			P3 coefficient for Q0 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x36C2	15:0	0x0000	P_GR_P3Q1 (R/W)				
			P3 coefficient for Q1 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x36C4	15:0	0x0000	P_GR_P3Q2 (R/W)				
			P3 coefficient for Q2 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x36C6	15:0	0x0000	P_GR_P3Q3 (R/W)				
			P3 coefficient for Q3 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x36C8	15:0	0x0000	P_GR_P3Q4 (R/W)				
			P3 coefficient for Q4 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x36CA	15:0	0x0000	P_RD_P3Q0 (R/W)				
			P3 coefficient for Q0 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x36CC	15:0	0x0000	P_RD_P3Q1 (R/W)				
			P3 coefficient for Q1 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x36CE	15:0	0x0000	P_RD_P3Q2 (R/W)				
			P3 coefficient for Q2 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x36D0	15:0	0x0000	P_RD_P3Q3 (R/W)				
			P3 coefficient for Q3 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x36D2	15:0	0x0000	P_RD_P3Q4 (R/W)				
			P3 coefficient for Q4 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x36D4	15:0	0x0000	P_BL_P3Q0 (R/W)				
			P3 coefficient for Q0 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x36D6	15:0	0x0000	P_BL_P3Q1 (R/W)				
			P3 coefficient for Q1 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x36D8	15:0	0x0000	P_BL_P3Q2 (R/W)				
			P3 coefficient for Q2 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x36DA	15:0	0x0000	P_BL_P3Q3 (R/W)				
			P3 coefficient for Q3 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x36DC	15:0	0x0000	P_BL_P3Q4 (R/W)				
			P3 coefficient for Q4 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x36DE	15:0	0x0000	P_GB_P3Q0 (R/W)				
			P3 coefficient for Q0 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x36E0	15:0	0x0000	P_GB_P3Q1 (R/W)				
			P3 coefficient for Q1 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x36E2	15:0	0x0000	P_GB_P3Q2 (R/W)				
			P3 coefficient for Q2 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x36E4	15:0	0x0000	P_GB_P3Q3 (R/W)				
			P3 coefficient for Q3 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x36E6	15:0	0x0000	P_GB_P3Q4 (R/W)				
			P3 coefficient for Q4 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3700	15:0	0x0000	P_GR_P4Q0 (R/W)				
			P4 coefficient for Q0 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3702	15:0	0x0000	P_GR_P4Q1 (R/W)				
			P4 coefficient for Q1 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3704	15:0	0x0000	P_GR_P4Q2 (R/W)				
			P4 coefficient for Q2 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3706	15:0	0x0000	P_GR_P4Q3 (R/W)				
			P4 coefficient for Q3 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x3708	15:0	0x0000	P_GR_P4Q4 (R/W)				
			P4 coefficient for Q4 for Gr. P_GR_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gr pixels.				
R0x370A	15:0	0x0000	P_RD_P4Q0 (R/W)				
			P4 coefficient for Q0 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x370C	15:0	0x0000	P_RD_P4Q1 (R/W)				
			P4 coefficient for Q1 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x370E	15:0	0x0000	P_RD_P4Q2 (R/W)				
			P4 coefficient for Q2 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3710	15:0	0x0000	P_RD_P4Q3 (R/W)				
			P4 coefficient for Q3 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3712	15:0	0x0000	P_RD_P4Q4 (R/W)				
			P4 coefficient for Q4 for Rd. P_RD_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Rd pixels.				
R0x3714	15:0	0x0000	P_BL_P4Q0 (R/W)				
			P4 coefficient for Q0 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x3716	15:0	0x0000	P_BL_P4Q1 (R/W)				
			P4 coefficient for Q1 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x3718	15:0	0x0000	P_BL_P4Q2 (R/W)				
			P4 coefficient for Q2 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x371A	15:0	0x0000	P_BL_P4Q3 (R/W)				
			P4 coefficient for Q3 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x371C	15:0	0x0000	P_BL_P4Q4 (R/W)				
			P4 coefficient for Q4 for Bl. P_BL_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Bl pixels.				
R0x371E	15:0	0x0000	P_GB_P4Q0 (R/W)				
			P4 coefficient for Q0 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3720	15:0	0x0000	P_GB_P4Q1 (R/W)				
			P4 coefficient for Q1 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3722	15:0	0x0000	P_GB_P4Q2 (R/W)				
			P4 coefficient for Q2 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x3724	15:0	0x0000	P_GB_P4Q3 (R/W)				
			P4 coefficient for Q3 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3726	15:0	0x0000	P_GB_P4Q4 (R/W)				
			P4 coefficient for Q4 for Gb. P_GB_PpQq registers are read successively when the row polynomial (Q) coefficients are calculated during the horizontal blanking period before a row containing Gb pixels.				
R0x3780	15:0	0x0000	POLY_SC_ENABLE (R/W)				
	15	0x0000	ENABLE 0: Lens Shading Disabled. 1: Lens Shading Enabled.				
	14:0	X	Reserved				
			The POLY_SC_EN bit in this register controls the lens shading correction function.				
R0x3782	15:0	0x03C4	POLY_ORIGIN_C (R/W)				
			Origin of polynomial function: applied as offset to X (col) coordinate of pixel.				
R0x3784	15:0	0x025C	POLY_ORIGIN_R (R/W)				
			Origin of polynomial function: applied as offset to Y (row) coordinate of pixel.				
R0x3786	15:0	0x0006	DIGITAL_CTRL_1 (R/W)				
	15:10	X	Reserved				
	9	0x0000	Reserved				
	8	0x0000	Reserved				
	7	0x000	Reserved				
	6	0x0000	Reserved				
	5	0x0000	LSB_ALIGN_PARA_OUT Align the pixel data LSB in parallel mode				
	4	0x0000	USE_1FRAME_SYNCED Both analog and digital gains will be effective at N+1 frame instead of N+2 frame				
	3	0x0000	PLL_TEST_MODE To send pll_tc on SHUTTER pad				
	2	0x0001	Reserved				
	1	0x0001	Reserved				
	0	0x0000	DRIVE_PIX_CLK PCLK tri-state control				
			digital_ctrl_1				

Table 4. MANUFACTURER SPECIFIC LIMITS

R/W (Read or Write) bit; RO (Read Only) bit; WO (Write Only) bit

Register Hex	Bits	Default	Name	Attributes			
R0x37C0	15:0	0x0000	P_GR_Q5 (R/W)				
	Parameter for parabolic roll-off algorithm for greenR pixels.						
R0x37C2	15:0	0x0000	P_RD_Q5 (R/W)				
	Parameter for parabolic roll-off algorithm for red pixels.						
R0x37C4	15:0	0x0000	P_BL_Q5 (R/W)				
	Parameter for parabolic roll-off algorithm for blue pixels.						
R0x37C6	15:0	0x0000	P_GB_Q5 (R/W)				
	Parameter for parabolic roll-off algorithm for greenB pixels.						

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